Tensor computations

Classification of perfect tensor formats with generic unique decomposition

Difficulty: extreme **Importance:** interesting to the community

Status: Partially resolved

Rating rationale: Extreme because excluding every further perfect identifiable format is a global classification problem; community importance comes from determining when generic tensor factors are uniquely recoverable at the dimension threshold.

Statement

Let $d \geq 3$, $n_1, \dots, n_d \geq 2$, and suppose the dimension-count value

$$r_* = \frac{\prod_{i=1}^{d} n_i}{1 + \sum_{i=1}^{d}(n_i - 1)}$$

is an integer. Such a format is called perfect. The conjecture asserts that a generic tensor in $\bigotimes_{i=1}^{d} \mathbb{C}^{n_i}$ has a unique minimal-length decomposition as a sum of rank-one tensors if and only if, after permuting factors, the format is

$$(2, k, k)\ (k \geq 2), \qquad (3, 4, 5), \qquad (2, 2, 2, 3).$$

Generic means on a nonempty Zariski-open subset of the entire tensor space. Rank one means $v_1 \otimes \cdots \otimes v_d$ with nonzero complex factors. Two decompositions are the same when their rank-one tensor summands agree after permutation; reciprocal rescalings within a summand create no new decomposition. No symmetry constraints are imposed.

Relevance

This determines whether a decomposition algorithm can uniquely recover components of a generic tensor at the parameter-count threshold. The exceptional positive cases support explicit decomposition algorithms.

References

1. J. D. Hauenstein, L. Oeding, G. Ottaviani, and A. J. Sommese, *Homotopy techniques for tensor decomposition and perfect identifiability*, J. reine angew. Math. 753 (2019), 1–22. DOI; primary preprint, §1, equation (2) and Conjecture 1.3, p.3; Theorems 1.1–1.2 establish the two sporadic positive cases.
2. A. Massarenti, M. Mella, and G. Staglianò, *Effective identifiability criteria for tensors and polynomials*, J. Symbolic Comput. 87 (2018), 227–237. Author PDF, §1, discussion of the Hauenstein et al. generic-identifiability conjecture.

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Rechecked Hauenstein et al., Theorems 1.1–1.2 and Conjecture 1.3, and searched for later perfect-format classifications. The listed positive cases are proved, while exhaustiveness of the nonsymmetric list remains conjectural. The solved symmetric analogue does not imply this claim. No full resolution was located; the 2019 publication remains the latest explicit source checked for this exact classification.